

Rapport sur le feedback du corps estudiantin

Student Feedback Report

Cours / Course :

Dynamic finite element analysis of structures

ME-473_SP25

SP25

Enseignant·e(s) / Teacher(s) :

Stefano Francesco Burzio

Centre d'appui à l'enseignement / Teaching Support Centre

15.07.2025



Dynamic finite element analysis of structures (ME-473_SP25)

EPFL

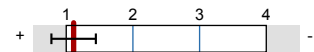
Stefano Francesco Burzio

- SP25

Nb réponses = 9/15 (60%)

Indicateurs globaux

Dans l'ensemble, je trouve que ce cours est bon.

m.=1,1
s.=0,3

Résultats des questions prédéfinies

1. Ce questionnaire permettra à la personne ou l'équipe qui enseigne, ainsi qu'à la section, de connaître votre appréciation du cours.

1.1) Merci d'indiquer votre section :

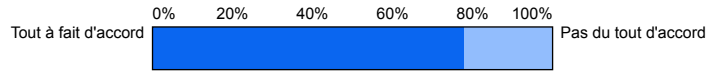
GM 100%

n=7

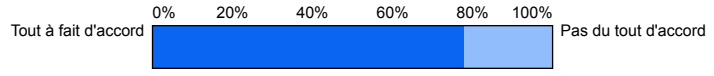
Merci d'évaluer les affirmations suivantes. / Please rate the following statements.

Tout à fait d'accord - D'accord - Pas d'accord - Pas du tout d'accord
 Strongly agree - Agree - Disagree - Strongly disagree

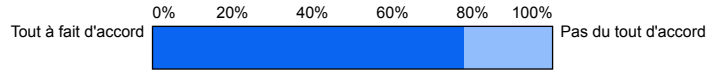
1.2) J'estime que cet enseignement est intéressant.

n=9
m.=1,2
md.=1
s.=0,4

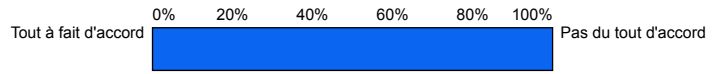
1.3) Je trouve que cet enseignement est bien organisé.

n=9
m.=1,2
md.=1
s.=0,4

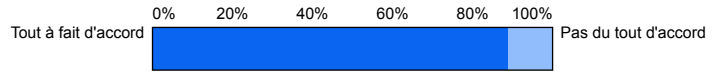
1.4) Je voyais clairement pendant le semestre ce que je devais connaître et être capable de faire au terme de cet enseignement.

n=9
m.=1,2
md.=1
s.=0,4

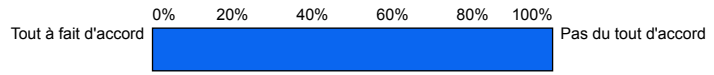
1.5) Les travaux demandés (exercices, TP, projets, lectures, etc.) m'ont permis d'assimiler la matière.

n=9
m.=1
md.=1
s.=0

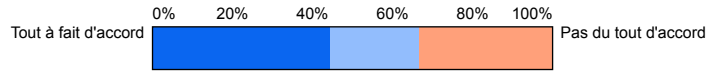
1.6) Pendant le semestre, je pouvais obtenir les conseils et le feedback nécessaires à mon apprentissage.

n=9
m.=1,1
md.=1
s.=0,3

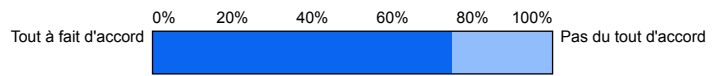
1.7) Je trouve que le climat de la classe est propice pour participer, intervenir et poser des questions.

n=9
m.=1
md.=1
s.=0

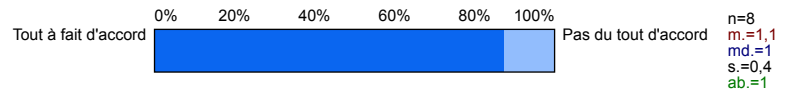
1.8) Je trouve que la charge de travail est en bonne correspondance avec les crédits/coefficients accordés au cours.

n=9
m.=1,9
md.=2
s.=0,9

1.9) Le cours est bien intégré et fait sens dans le cursus bachelor/master.

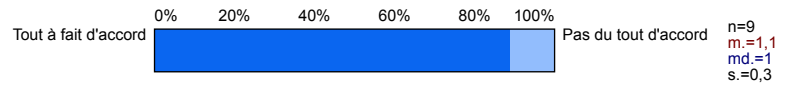
n=8
m.=1,3
md.=1
s.=0,5
ab.=1

1.10) Je trouve que l'encadrement des séances d'exercices est satisfaisant, tant en qualité qu'en nombre d'assistants.



2. Merci de donner votre appréciation générale et vos commentaires sur le cours

2.1) Dans l'ensemble, je trouve que ce cours est bon.



Résultats des questions ouvertes

2. Merci de donner votre appréciation générale et vos commentaires sur le cours

2.2) Merci de faire part de vos commentaires sur ce cours :

- Cours super intéressant.
Les exercices permettent de bien comprendre la théorie et la physique derrière les équations.
Les mini project permettent de mettre en pratique la théorie et aussi de mieux comprendre la théorie.
Cependant, j'ai l'impression que les éléments vu en cours restent assez basique (beam et plates principalement) et permet donc plus difficilement d'étudier des systèmes plus complexes (en 3D également).
De plus, utiliser des softwares d'élément finis serait un vrai plus (peut être garder 2 projets sur matlab et un projet sur Abaqus ou autre).
Finalement, la charge de travail est un peu élevé pour 3 crédits, car les projets en plus des séries d'exercices requiert pas mal de temps, sans compter qu'il faut encore bien réviser et se préparer à l'examen final en plus. Je pense que rajouter un crédit et rendre les projets un peu plus long (peut être un peu plus complexe que juste un beam ou juste une plate) et/ou diminuer le nombre de personne par groupe 2-3 max (car c'est pas facile de bien se répartir les tâches quand on est jusqu'à 5) pourrait être une solution. Par ailleurs, je trouve que les rapport devrait être plus concis en terme de contenu, je pense qu'expliquer la méthodologies et la théorie est un peu superflu puisque de toute manière la théorie devra être connue pour l'examen. Un focus sur les résultats et discussion serait plus intéressant. Par ailleurs, je pense pas non plus que le code doit être évalué en détails.
- Interesting and well structured course, but a bit to mathematical for my taste. As the focus of finite element in engineering after graduation, is often the practical use of the method, I think it would be more benefiscal/usefull to focus more on the practical implementation.

Also quite a lot of time was used on covering different elements, where it could instead have been more interesting to use more time on the later topics of the course, and maybe to cover/introduce how to approach advanced topics such as contact modelling.

Laslty I would recommend to rethink whether it is necessary to have three projects and an exam for a 3 credit course. Due to the extent asked for in the projects (such as including a section on theory), I would for example not think it is necessary with an exam as well. Also due to the general practicality of the field, I think projects could serve as the main evaluation method.
- I think it was one of the best courses I had at EPFL. Nevertheless, I'd like to provide some suggestions:
- 3 projects plus exams for 3 credits, I think it's too much. I believe it would be interesting to maybe have just the project and have one completely focused on a software rather than just coding from scratch in MATLAB. Otherwise, it would be interesting to have a small midsemester exam, rather than one in the exam session.
- It might be interesting to span over more arguments rather than so many different element types. I'd personally prefer to know more about non-linear analyses for instance
- The slides are amazing, super clear and professional, but few errors (pointed out during the semester) undermined clarity.
- The mini-project grading was very strict, completely different from different course I add. I don't think it is a problem by itself, because it pushed me to work more and better, but at least for me, it wasn't clear from the start of the semester. Indeed I'm not fully satisfied by some of the grades I received, as I would have done differently if I knew before
- At last, I'd like to have some more practical example, that's why I believe a bigger final project rather than an exam might be useful
Overall, compliments to the professor and TA.
- The class is interesting, but the organization of the slides could use refinement. Although they are clearly laid out visually, the notation becomes quickly confusing and sometimes many steps are skipped between slides while in others each step is spelled out, so it can be confusing to know at what level we should be studying the concepts. The projects were interesting and applicable, but in my opinion a lot of work considering the percentage of the grade they are worth, and that we also have a final exam. I would have liked one of the projects to involve actual simulation software, as Matlab is not a tool really adapted to FEM analysis, beyond showing basic implementation. The first 2/3 of the course I thought were well organized, but the last 1/3 I was a bit lost as to the application of what we were doing. It would have been nice to discuss in the class at the end a bit how these concepts could be applied to more complex structures, as we stayed looking at just plates and beams. Professor very accessible and obviously cares about/attentive to the students learning. Always tried to answer questions to the best of his abilities and would look into questions he was not familiar with.
- this course is excellent, but we felt it the first time it was being given
the teacher did not always know what was on his slides (beautiful slides however!) , i think because some images or notations were inherited from the previous course
i think it would be better to keep clear of the old one and build a course more modern and unrelated to the previous one

maybe find a better notation as well, please lets remove indices or subscripts from the left side of a letter, its horrendous

the projects were very interesting, but im not sure their format is adapted to the course:
three possible work structures options, each with drawbacks:
1) one person codes and writes the results, the rest write the other report sections --> only one person actually applied the course knowledge
2) everyone writes on one code together, and joins for the report report --> no one likes coding with several people on one computer, or if dividing the code among the group, adapting their notation to the others
3) everyone writes their separate code and join for the report, which means everyone got to apply the course knowledge --> very inefficient in terms of total hours worked
a possible solutions would be to work on the projects by pairs, as it is more practical to work on code being only two people, but then lighten the report load, ie make it more results and discussion driven and less academical with intro/theory/results/discussion/ conclusion. its not that useful to include to theory since we just copy it from the slides

also, given the time and effort invested in the reports, the grading was harsh. we are mechanical engineering students, not physics students on their way to a phd, removing points for axis labels or log scales is absurd. please either grade it accordingly or publish the expected form the results should take before project 1

since some exercise sessions were dedicated to the mini projects, we did not get to train every subject in the course, maybe some revision exercises with these subjects could be put on moodle?